VIRTUAL UNIVERSITY OF PAKISTAN

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
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(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.

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Best of luck!

PAST PAPER

Question No : 1 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

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Answer (Please select your correct option)

- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\text{-------}\frac{5}{6}\frac{3}{4}\frac{1}{2}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\text{-------}\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\text{-------}\frac{6}{7}\frac{4}{5}\frac{2}{3}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\text{-------}\frac{6}{5}\frac{4}{3}\frac{2}{1}$

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = 4\cos 3x$?

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Answer (Please select your correct option)

- $\frac{\pi}{4}$
- $\frac{\pi}{3}$
- $\frac{2\pi}{3}$ ⇦
- π

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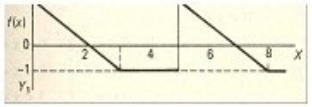
Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Match the following periodic function with its graph.

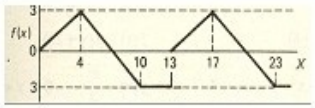
$$f(x) = \begin{cases} 2 - x & 0 < x < 3 \\ -1 & 3 < x < 5 \end{cases}$$

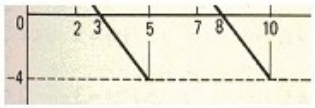
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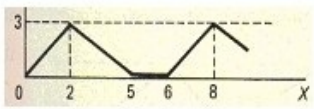
Answer (Please select your correct option)



correct





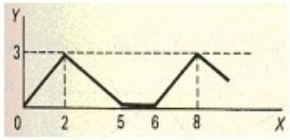


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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



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Answer (Please select your correct option)

- 2
- 5
- 6
- 8

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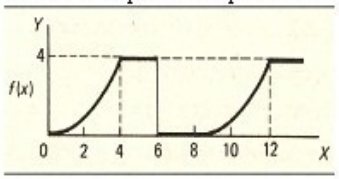
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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



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Answer (Please select your correct option)

- 0
- 4
- 6
- 8

correct

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

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Answer (Please select your correct option)

- Line integral of $\vec{F}$ along a curve, depends only on the endpoints of that curve, not on the particular route taken.
- Divergence of $\vec{F}$ should be zero
- Gradient of $\vec{F}$ should be zero.
- $\vec{F} = 0$

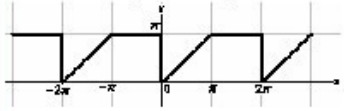
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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



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Answer (Please select your correct option)

- π
- $-\pi$ correct
- 2π
- -2π

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Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Let L denotes the Laplace Transform.
According to First Shift Theorem, if $L\{F(t)\} = f(s)$ then which of the following equation holds?
s *and* a are constants.

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Answer (Please select your correct option)

- $L\{e^{-at}F(t)\} = f(s+a)$
- $L\{e^{-at}F(t)\} = f(s)$
- $L\{e^{-at}F(t)\} = f(a)$
- $L\{e^{-at}F(t)\} = f(s-a)$

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^2 \cos 2x$ is ----------

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Answer (Please select your correct option)

- Neither even nor odd
- Odd function
- Even function

correct

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an even function is symmetrical about ----------

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Answer (Please select your correct option)

- x-axis
- y-axis
- origin

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s) = \frac{3}{s-2} - \frac{2}{s}$?

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Answer (Please select your correct option)

- $3te^{2t} - 2$
- $3e^{2t} - 2t$
- $3e^{2t} - 2$ correct
- None of these.

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

Answer (Please select your correct option)

- $L\{\sin 3t\} = \frac{3}{s^2+9}$ correct
- $L\{\sin 3t\} = \frac{s}{s^2+9}$
- $L\{\sin 3t\} = \frac{1}{s-3}$
- $L\{\sin 3t\} = \frac{3!}{s^4}$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If L denotes laplace transform then
$L(te^{5t}) =$

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Answer (Please select your correct option)

- $L(te^{5t}) = \frac{1}{s^2 - 5}$
- $L(te^{5t}) = \frac{1}{s^2 + 5}$
- $L(te^{5t}) = \frac{1}{(s+5)^2}$
- $L(te^{5t}) = \frac{1}{(s-5)^2}$ correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

What is $L(-6)$ if L denotes Laplace Transform?

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Answer (Please select your correct option)

- $L(-6) = \frac{1}{s+6}$
- $L(-6) = \frac{-6}{s}$ correct
- $L(-6) = \frac{s}{s^2+36}$
- $L(-6) = \frac{-6}{s^2+36}$

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector field is a vector each of whose components is a scalar field

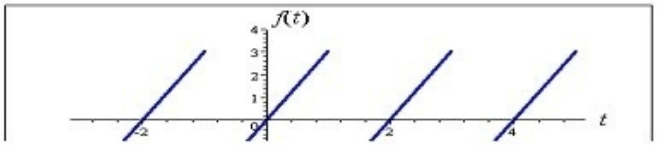
Answer (Please select your correct option)

- True
- False

correct

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Made by: Waqar Siddhu

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)



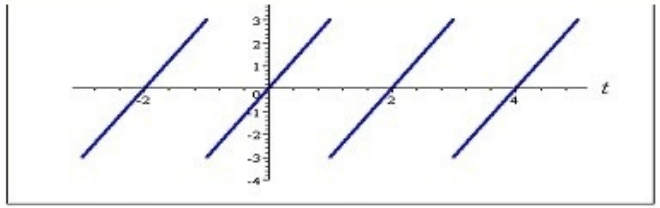
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Answer (Please select your correct option)

- An odd function
- An even function
- Neither even nor odd

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)



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Answer (Please select your correct option)

- An odd function
- An even function
- Neither even nor odd

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)



The graph of "saw tooth wave" given above is ----------

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Answer (Please select your correct option)

- An odd function
- An even function
- Neither even nor odd

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

All axes are positive in ---------- octant.

Answer (Please select your correct option)

- First
- Second
- Fourth
- Eighth

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

The angles which a line makes with positive x ,y and z-axes are known as ----------

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Answer (Please select your correct option)

- Direction cosines
- Direction ratios
- Direction angles

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

If α, β and γ are the direction angles of a line then

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma =$

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Answer (Please select your correct option)

- 0
- 1 correct
- 2
- T

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Intersection of two non parallel planes is a . . . . . .

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Answer (Please select your correct option)

- Plane
- Point
- Straight line
- None of these.

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

$$\lim_{(x,y)\to(0,1)} \frac{xy^2}{x^2+y^2} =$$

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Answer (Please select your correct option)

- ∞
- 0
- 1
- 0.5

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $z = \frac{1}{\sqrt{x+y}}$ is discontinuous at origin because at the point (0,0), it . . . . . .

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Answer (Please select your correct option)

- has a jump
- has a hole
- approaches towards infinity
- approaches towards zero

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Gradient of a scalar function always results in a . . . . . function.

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Answer (Please select your correct option)

- Scalar
- Continuous
- Vector
- Constant

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

At the critical point (x_0, y_0), if $D = f_{xx}(x_0y_0)f_{yy}(x_0y_0) - f_{xy}^2(x_0y_0) < 0$ then f has a . . . . . .

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Answer (Please select your correct option)

- Relative maximum at (x_0, y_0)
- Relative minimum at (x_0, y_0)
- Saddle point at (x_0, y_0)
- No conclusion can be drawn

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

The volume of parallelepiped with dimensions x, y, z is . . . . . .

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Answer (Please select your correct option)

- $V = x^2y^2z^2$
- $V = x + y + z$
- $V = \sqrt{xyz}$
- $V = xyz$

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) : 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$, then $\iint_R (6x^2 + 4xy^3)dA =$ . . . . . .

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Answer (Please select your correct option)

- $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$
- $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$
- $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$ correct
- $\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

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Answer (Please select your correct option)

- $\left(7, \frac{3\pi}{4}\right)$ correct
- $\left(-7, \frac{3\pi}{4}\right)$
- $\left(-7, \frac{-\pi}{4}\right)$
- $\left(7, \frac{-3\pi}{4}\right)$

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

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Answer (Please select your correct option)

- $\left(-2, \frac{\pi}{2}\right)$
- $\left(-2, \frac{\pi}{4}\right)$
- $\left(-2, \frac{3\pi}{4}\right)$
- $\left(-2, \frac{\pi}{3}\right)$

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about . . . . . .

Answer (Please select your correct option)

- y-axis
- Pole
- Initial line
- origin

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

In polar coordinate system, x- axis is also called . . . . . .

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Answer (Please select your correct option)

- Polar axis correct
- Pole
- Imaginary axis
- None of these

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

The point $p(0,\theta)$ in polar coordinate system lies on . . . . . .

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Answer (Please select your correct option)

- Polar axis
- y-axis
- Pole
- None of these

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become . . . . . where $a \le \theta \le b$ and $c \le r \le d$.

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Answer (Please select your correct option)

- $\int_a^b \int_c^d f(r,\theta)rdrd\theta$
- $\int_a^b \int_c^d f(r,\theta)drd\theta$ correct
- $\int_a^b \int_c^d f(r,\theta)rd\theta dr$
- $\int_a^b \int_c^d f(r,\theta)d\theta dr$

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The parametric equations that correspond to the vector equation $\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$ are . . . . . .

Answer (Please select your correct option)

- $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 0$
- $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \quad z = 0$
- $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 1$
- $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\vec{R}(t)$ is the anti-derivative of a given vector valued function $\vec{r}(t)$ i.e., $\vec{R}'(t) = \vec{r}(t)$ then $\int_a^b \vec{r}(t)dt = \ldots\ldots\ldots$

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Answer (Please select your correct option)

- $\vec{R}(a) + \vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a) + c$ where c is a non-zero constant.
- $\vec{R}(a).\vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a)$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t)\times\vec{r}_2(t)]$ =..... where "×" and "." represent the cross and dot product respectively.

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Answer (Please select your correct option)

- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \vec{r}_2 \times \frac{d\vec{r}_1}{dt}$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

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Question No : 36 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A vector valued function $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ is continuous if . . . . . .

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Answer (Please select your correct option)

- Atleast one of its components is continuous.
- All of its components are necessarily differentiable.
- All of its components are continuous.
- Limit exists for all of its components.

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Question No : 37 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Given a vector valued function $\vec{r}(t) = \frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ and its anti-derivative $\vec{R}(t) = \ln(t-3)\hat{i} + e^t\hat{j}$, then $\int \vec{r}(t)dt = \ldots\ldots\ldots\ldots$

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Answer (Please select your correct option)

- $\ln(t-3)\hat{i} + e^t\hat{j} + c$
- $(t-3)\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $(t-3)^{-1}\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $\frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ ⇦

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

For any two vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t).\vec{r}_2(t)] =$ ..... where "×" and "·" represent the cross and dot product respectively.

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Answer (Please select your correct option)

- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} . \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\vec{r}_1 . \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \times \vec{r}_2$ ⇦

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A single curve can be represented by . . . . . vector valued function(s).

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Answer (Please select your correct option)

- Two
- Infinitely many
- Single ⇦ correct
- Three

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

A function is said to be smooth if it's derivative is . . . . . on any value of its domain.

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Answer (Please select your correct option)

- continuous and non zero
- piecewise continuous
- defined and non zero
- differentiable

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \, dx$

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Answer (Please click here to Add Answer)

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^3 + x^2$

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Answer (Please click here to Add Answer)

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Consider the point $\left(\frac{\sqrt{3}}{2}, \sqrt{3}, \frac{1}{2}\right)$ in rectangular coordinate system. Find the value of " ρ " in spherical coordinates.

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Answer (Please click here to Add Answer)

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down the expression for the arc- length of the curve represented by the vector valued function $\vec{r}(t) = \cos t^2 \hat{i} + \sin t^2 \hat{j}$ where $0 \le t \le \frac{\pi}{2}$. (Do not evaluate the expression).

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Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis cosine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\cos^3 x + \cos^4 x\right) dx$

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Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Prove whether the following function is even, odd or neither.

$f(x) = \sin^2 x \cos 3x$

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Answer (Please click here to Add Answer)



Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Find the region where the function $f(x, y) = \sqrt{9 - x^2 - y^2}$ is continuous.

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Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = (4+3t)\hat{i} + (2-2t)\hat{j}$ when $3 \le t \le 4$?

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Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine the Fourier co-efficient b_n of the periodic function defined below:

$f(x) = 2x + 1 \qquad 0 < x < 2$

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Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\hat{j} + xz^2\hat{k}$

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Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Using second partial derivative test show that the function $f(x, y) = xy(24 - x - y)$ is maximum at $x = 8, y = 8$.

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Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Let $\vec{r}(t) = t^2\,\hat{i} + t\,\hat{j} + (t^2 - 5)\,\hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r}'(t)$ are perpendicular to each other.

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Answer (Please click here to Add Answer)

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